Student success in academics based on enrollment demographics

Are you able to predict whether a student will drop out, stay enrolled or graduate based on academic, socioeconomic and demographic information?

Dataset:

Predict Students' Dropout and Academic Success

Realinho, V., Vieira Martins, M., Machado, J., & Baptista, L. (2021). Predict Students' Dropout and Academic Success [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5MC89.

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Variables and Inputs:

The target variable is simply “target” and has three categories (dropout, enrolled, and graduate). This category is the category the student is in at the ***end*** of the ***normal*** course duration (e.g. 4 years)

The list of main input features is:

1) Marital Status

2) Application order (e.g. was this their first-choice university or last choice?)

3) Course

4) Daytime/evening attendance

5) Previous qualification

6) Previous qualification grades

7) Nationality

8) Father/Mother's qualification

9) Father/Mother's occupation

10) Admission Grade

11) Gender

12) Scholarship Holder

13) Age at enrollment

14) International

These are a subset of all available features. Note that the data for these features is gathered ***on enrollment***, not at end of course duration like the target variable.

Records and Data Quality:

The data is 4424 instances long, with each instance being one student. The data was gathered at a Portuguese university. The data has been preprocessed to handle anomalies, unexplainable outliers, and missing values. However, some data might have to be grouped for ease of analysis. For example, instead a list of all various nationalities, many of which may only have a small handful of records, we could instead group them into categories such as Native Portuguese, from another Portuguese speaking country, non-Portugal EU country, and miscellaneous.

Baseline, Candidate Models and Evaluation Metrics:

The proposed Baseline for this project is a dummy baseline, simply guessing the most common answer. Potential candidate models that could be used are Decision Trees, Multiclass Support Vector Machines, and logistical regression. The evaluation metrics I will be focusing on are accuracy, precision, recall, and F-Scores (specifically F1, to balance precision and recall equally).

Privacy and other Machine Learning Risks:

It is very important that this project respects privacy and fairness and prevents misuse of the model. The single most important thing is to ensure that all work done complies with

the General Data Protection Regulation (GDPR). Specifically, it's important to prevent specific students from being identified. It's also important to remember that, if this model is applied to new students, that we understand that predictions are just predictions, and not a set in stone fact about a student's future.

Link to Github Repository:

https://github.com/EoinCorless/CT4101_Student_dropout_and_success